GEO Monitoring Service

First version (pilot) architecture on AWS

Document	Version	Status	Owner	Date
First version architecture	0.1	Pilot	Rui, JiLai AI Tech	12 Oct 2026

1. Goal

Run the GEO monitoring service for the first 1-3 agency clients at the lowest cost and operational effort, while keeping the same application design as the target architecture. Everything is defined in Terraform in `codes/geo-pilot-aws`.

- One EC2 server runs everything in Docker: Caddy (HTTPS), the web app and the worker.
- No load balancer and no NAT gateway, which are the largest fixed costs at this scale.
- Workers read jobs from SQS from day one, so moving to the target architecture needs no code change.

2. Architecture

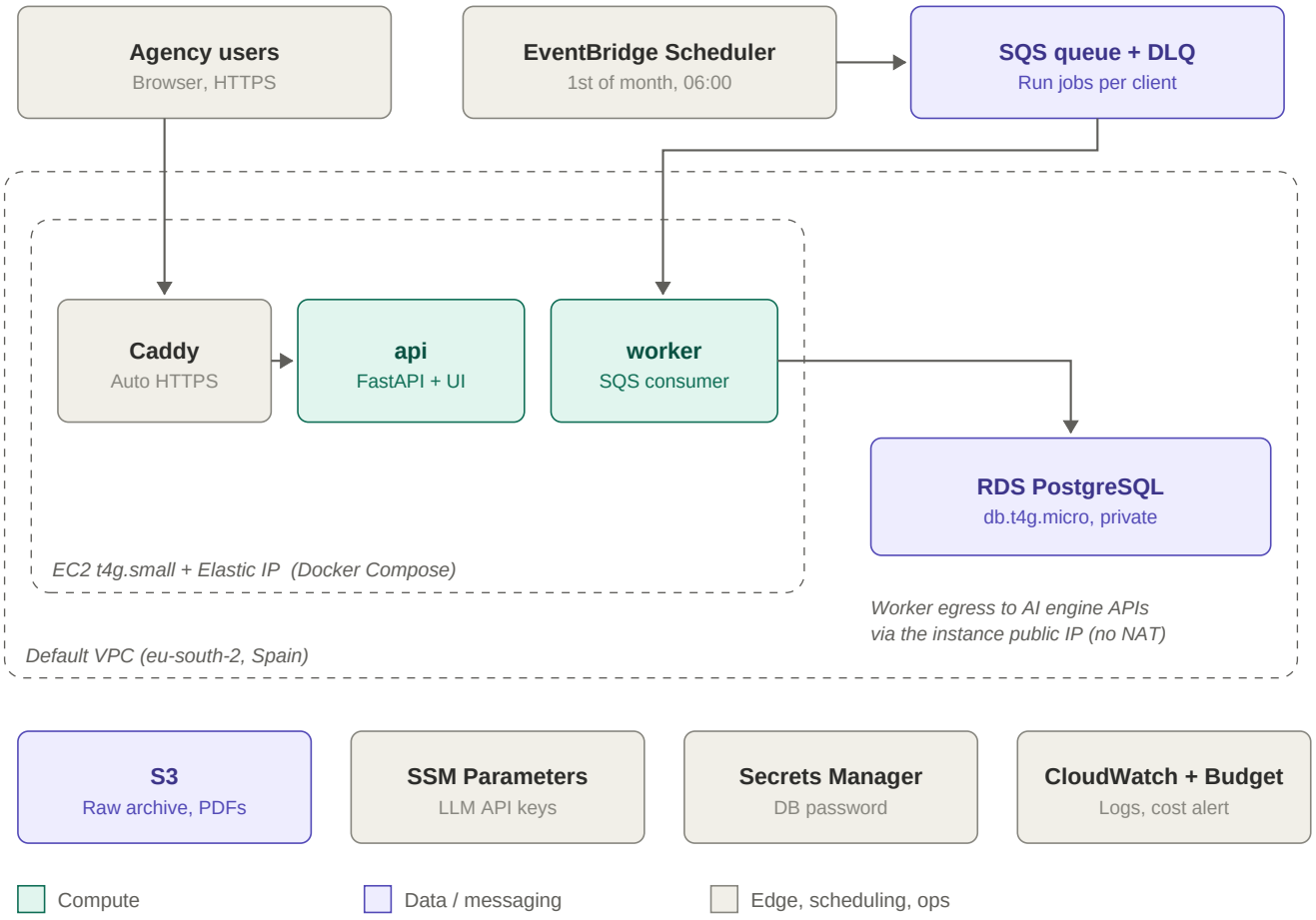

Figure 1. First version on AWS

3. Components

Resource	Configuration	Purpose
EC2	t4g.small (ARM), Amazon Linux 2023, 30 GB gp3 encrypted, Elastic IP, IMDSv2	Runs Docker Compose: caddy, api, worker
Caddy	Automatic Let's Encrypt certificate for the dashboard domain	HTTPS without ALB/ACM
api container	FastAPI backend + dashboard	Agency UI, on-demand runs
worker container	Long-polls SQS, calls AI engines, analyzes, writes results	Monthly and on-demand runs
RDS PostgreSQL 16	db.t4g.micro, 20 GB gp3 (autoscaling to 100 GB), single-AZ, private, 7-day backups, deletion protection	Configs, answers, metrics, tasks
SQS	Jobs queue + DLQ (3 attempts), SSE, long polling, 900 s visibility	Job queue
EventBridge Scheduler	cron(0 6 1 * ? *), Europe/Madrid	Monthly run trigger
S3	Private, encrypted, raw/ to Glacier IR after 180 days	Raw archive, PDF reports
SSM Parameter Store	One SecureString per provider: openai, anthropic, perplexity, gemini	LLM API keys
Secrets Manager	RDS-managed master password	DB credentials
CloudWatch Logs	/geo-pilot/app, 30-day retention, Docker awslogs driver	Logs
AWS Budgets	Email at 80% forecast and 100% actual of monthly limit	Cost guard

4. Network and security

- Uses the account's default VPC. The EC2 security group allows only TCP 80 and 443 inbound; all outbound (AI engine APIs, AWS services).
- RDS is not publicly accessible; its security group allows port 5432 only from the EC2 security group.
- No SSH key and no port 22: shell access through SSM Session Manager.
- The instance role grants least-privilege access to the queue, the bucket, the /geo-pilot/ SSM parameters, the DB secret and the log group. Containers use it through IMDSv2 (hop limit 2).
- API keys are set with the AWS CLI and never stored in Terraform state, files or images.

5. Run flow

Message	Sent by	Worker action
{"type": "monthly_run"}	EventBridge Scheduler, 1st of month	Enqueue one client_run per active client
{"type": "client_run", "client_id": ...}	Worker, or dashboard for on-demand	Run the client's prompts on all engines (3 samples), analyze, store, build report, delete message
(failed 3 times)	SQS redrive	Message lands in the DLQ for inspection

6. Deployment steps

- **Infrastructure:** in `terraform/`, copy terraform.tfvars.example, set alert_email, run `terraform init` and `terraform apply` (about 10 minutes).
- **API keys:** `aws ssm put-parameter --type SecureString --overwrite --name /geo-pilot/openai_api_key --value ...` for each provider.
- **DNS:** A record for the dashboard domain to the public_ip output.
- **App:** open a shell with `aws ssm start-session --target <instance_id>`, put the deploy/ files and the app code in /opt/geo, create app.env, run `docker compose up -d --build`.

7. Cost estimate

Roughly USD 30-40 per month for AWS before AI engine API costs, dominated by EC2 t4g.small, RDS db.t4g.micro with storage, and the public IPv4 address. SQS, Scheduler, SSM, S3 and logs are cents at this scale. AI API usage is about 600 calls per client per month at the default settings. Verify with the AWS Pricing Calculator for eu-south-2.

8. Known limits and upgrade triggers

Limit in V1	Risk	Upgrade when
Single EC2 instance	Downtime during patching or failure; dashboard and runs stop	Agency needs an availability commitment
Worker shares the server with the web app	A large run can slow the dashboard	More than ~10 clients or long runs
RDS single-AZ	Restore from backup on AZ failure (minutes to hours)	Paying agencies, SLA
Public subnet for EC2	Larger attack surface than private subnets (mitigated by SG + no SSH)	Security review or enterprise client
No per-agency login service	Basic app-level auth only	Second agency onboarded (add Cognito)

Each upgrade is an infrastructure change only; see the services architecture document, section 10.